A Hierarchical Gamma Mixture Model Toward Hidden Markov Random Field for High-Resolution SAR Image Segmentation

Xue Shi and Yu Li

***Abstract*— Accurately modeling the distributions of spectral intensities is an effective way to obtain accurate segmentation results. Existing mixture models mostly fail to establish accurate statistical models of high-resolution SAR images due to the complex statistical features of their spectral intensities. We propose a hierarchical gamma mixture model (HGaMM)-based high-resolution SAR images segmentation algorithm. In the proposed algorithm, a statistical model for SAR images is built using HGaMM, which can model heavy-tailed, asymmetric, multimodal, or flat distributions with its hierarchical structure. The algorithm allows for the effective utilization of spectral intensity information required for image segmentation. The HGaMM consists of several components that serve as the first layer, and several elements under each component serve as the second layer. The component weight is constructed using posterior probabilities of local pixels aimed at reducing noise effects. The segmentation model can then be established by the Bayesian theorem. A new expectation maximization/adding or deleting Markov chain Monte Carlo (EM/ADMCMC) is incorporated to implement parameter estimation and determine the optimum number of components. Several experiments were implemented on simulated and real high-resolution SAR images to evaluate the applicability and efficiency of the proposed approach. The experimental results show that the proposed HGaMM algorithm outperforms traditional algorithms and can obtain accurate results.**

***Index Terms*— Bayesian theorem, expectation maximization (EM)/adding or deleting Markov chain Monte Carlo (MCMC), hidden Markov random field (MRF), hierarchical Gamma mixture model (HGaMM), SAR image segmentation.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active remote sensing system for Earth observation that provides images in all weather conditions at any given time. SAR has become a popular research topic in the remote sensing field and plays a critical role in resource management and military applications. Recently, the application of SAR images has diversified into various fields, including crop growth and disaster monitoring [1]–[3]. SAR image segmentation is highly significant to numerous applications, where accurate segmentation results are crucial in providing a wide variety of social production information [4]. However, speckle noise, generated by the coherent processing of radar signals, can severely affect the quality of SAR segmentation results. The problem becomes more severe for higher spatial resolution SAR images. Hence, designing segmentation algorithms that would resolve speckle noise, specifically in high-resolution SAR images, would be extremely useful for many applications.

In the last decade, researchers have proposed a number of segmentation algorithms for SAR images, such as threshold, statistical model, and cluster-based segmentation algorithms [5]–[10]. Some have proposed that the distribution of spectral intensities in SAR images satisfies particular statistical laws [11], [12]. Hence, segmentation algorithms based on statistical models have been used extensively. The main advantage of using statistically based segmentation models is that the information of the image can be easily modeled in a probabilistic way, such as using the intensity and location of pixels. In addition, the approach has strong statistical theoretical foundations. In recent years, many segmentation algorithms have been proposed that are based on the assumption that the probability distribution of spectral intensities in a class approximately follows Gamma distribution [13], [14]. To reduce the impact of speckle noise, researchers build the segmentation model that combines the regular tessellation or locally spatial relation of pixels [15]. However, these algorithms fail to take into account the difference of spectrum in the same region, which then leads to wrongly segmented results.

Finite mixture model (FMM) [16], [17] has been proposed to effectively build the statistical model of remote sensing images. In image segmentation, FMM is used to model the statistical distribution of spectral intensity with the weighted components. The likelihood of model parameters can be constructed as the statistical model. Image segmentation can then be transformed into parameter estimation using the maximum likelihood function [18]–[20]. Currently, the research on the algorithm has included building the accurate statistical model, estimating the optimal model parameters, and adaptively determining the number of components. In order to build an accurate statistical model, which is an important prerequisite

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to obtain accurate results, various mixture models have been designed for image segmentation.

However, modeling complex statistical distributions of spectral intensities in SAR images using existing FMMs can be complex and difficult, given that their statistical distribution can be heavy-tailed, asymmetric, flat, or even multimodal. For example, the Gaussian mixture model (GMM) has been used extensively in image segmentation and has a simple structure easy to implement. However, its symmetrical and unimodal features are not suitable for more complex statistical distribution models [21], [22]. Thus, GMM is mainly used in multispectral and optical image segmentation and is rarely used in segmentation for SAR images [23]–[26].

To address this limitation, researchers have proposed the Student's-t mixture model (SMM) to build the statistical model for SAR images [27], [28]. Its components can be heavy-tailed, and the thickness of the tail is controlled by its freedom parameter. SMM provides greater flexibility than the GMM in building the statistical model for SAR images [29]–[31]. However, modeling asymmetrical distribution may be difficult, given SMM's components.

Another model that has been proposed and used in SAR image segmentation is the Gamma mixture model (GaMM) [32]–[34], having asymmetric and heavy-tailed components [35]. Its main drawback is that its components are unimodal, which fails to address bimodal and multimodal distributions of spectral intensities. Moreover, its shape parameter contained in the Gamma function is difficult to estimate using maximum likelihood estimation. Besides, generalized Gamma mixture model (GGaMM) is also widely applied in SAR image segmentation, due to the form diversity and modeling flexibility of generalized Gamma distribution [36], [37]. Many distributions are the special cases of generalized Gamma distribution, such as exponential, Gamma, and Weibull distributions. And it has a better performance to model the asymmetric and heavy-tailed distribution than Gamma distribution. However, the problem of its parameters estimation limits the application of GGaMM in image segmentation. For that, many modified EM algorithms were proposed to estimate the parameters of GGaMM [38], [39]. But the derivation process is more complex and the number of parameters is larger in [39].

To model the complex distributions of spectral intensities, researchers have utilized the weighted sums of distributions instead of the component for image segmentation [40], [41]. However, their algorithms do not consider the spatial relation of pixels and are susceptible to image noise. Moreover, they are difficult to segment high-resolution SAR images, due to the diversity and complexity of objects and the uncertainty of imaging conditions. In addition, the above algorithms cannot solve the problem of adaptive determination of the component number.

To carry out image segmentation, a number of approaches using mixture models have been developed to estimate segmentation parameters. Expectation maximization (EM) is a well-known method to solve the statistical model containing the hidden variable. Its main advantage is that it estimates parameters in the closed form [42]. EM is more widely used to estimate GMM parameters compared with other mixture models [43], [44], since some parameters of complicated models cannot be estimated at M-step. For some instances, EM may have difficulty estimating the freedom parameter of SMM contained in Gamma function, such that EM fails to obtain the expression of the shape parameter of GaMM. To avoid this concern, the Markov chain Monte Carlo (MCMC) is utilized to optimize parameters through iterations [13]. The algorithm is suitable for parameter optimization in complicated models. To suitably adapt to the number of components, the reversible jump MCMC (RJMCMC), an improvement of the MCMC algorithm, has been proposed to sample parameters in different dimensions [45]–[47]. However, the main drawback of the RJMCMC algorithm is its low efficiency due to having a large number of samples.

To solve this problem and implement accurate segmentation, we propose a hierarchical Gamma mixture model (HGaMM)-based high-resolution SAR images segmentation algorithm. HGaMM is constructed by weighted components, like the traditional FMM, to establish accurate models of spectral intensities in each objective region and uses elements defined by Gamma distribution. The HGaMM hierarchy consists of the components in the first layer and the elements in the second layer. In the proposed algorithm, the statistical model based on HGaMM is built first. HGaMM can build an accurate statistical model of complex distributions and fully utilize the information of spectral intensities to improve accuracy. The proposed algorithm constructs the component weights incorporating the posterior probability of neighboring pixels by Markov random field (MRF), aimed at introducing the locally spatial relations of pixels. This limits the impact of image noise and simplifies the estimation of the component weight. The posterior distribution of model parameters can then be constructed as a segmentation model following the Bayesian theory. Finally, an EM/adding or deleting MCMC (EM/ADMCMC) is proposed to solve the segmentation model. This process can adaptively estimate the optimum number of components and avoid the limitations of MCMC and EM algorithms. Numerous experiments were implemented comparing the proposed algorithm with other segmentation approaches. The results show that the proposed HGaMM algorithm performed better segmentation of high-resolution SAR images.

This article is organized into four sections. In Section II, the proposed algorithm is described in detail, including the development of the HGaMM, the HGaMM-HMRF segmentation model, and model parameter estimation. The experiments and the results are discussed in Section III, and the conclusion is presented in Section IV.

II. Proposed Algorithm

A. Hierarchical Gamma Mixture Model

The HGaMM structure can be described as follows: Let $\boldsymbol{X} = \{X_i;\ i = 1, 2, \ldots, n\}$ be a random field and $\boldsymbol{S} = \{1, 2, \ldots, s\}$ be the state spaces of $\boldsymbol{X}$. For $\forall i$, we have $X_i \in \boldsymbol{S}$. The segmented image can be viewed as the set of spectral intensities. It presents $\boldsymbol{x} = \{x_i;\ i = 1, 2, \ldots, n\}$, which is a configuration of $\boldsymbol{X}$, and x_i is the intensity of pixel i.

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Let $\boldsymbol{Z} = \{Z_i;\ i = 1, 2, \ldots, n\}$ be a random field and its state spaces be $\boldsymbol{K} = \{1, 2, \ldots, k\}$. For $\forall i$, we have $Z_i \in \boldsymbol{K}$. Let $\boldsymbol{z} = \{z_i;\ i = 1, 2, \ldots, n\}$ denote the configuration of $\boldsymbol{Z}$. Image segmentation is used to assign n pixels into k objective regions. The segmentation result can then be viewed as a set of labels. Let $\boldsymbol{z}$ be the set of labels and z_i be the label of pixel i.

To build an accurate statistical model of spectral intensities for SAR images with high resolution, each objective region is divided into m subregions in which spectral intensities are approximately homogeneous. Let $\boldsymbol{Y} = \{Y_i;\ i = 1, 2, \ldots, n\}$ be a random field and the state spaces is $\boldsymbol{M} = \{1, 2, \ldots, m\}$. For. $\forall i$, we have $Y_i \in \boldsymbol{M}$. Let $\boldsymbol{y} = \{y_i; i = 1, 2, \ldots, n\}$ denote the configuration of $\boldsymbol{Y}$. And $\boldsymbol{y}$ presents the set of sublabels and y_i presents the sublabel of pixel i. The sublabel y_i is only meaningful if the label of pixel i is known. For example, when $z_i = c$ and $y_i = j$, pixel i is allocated to subregion j of objective region c. In a subregion, the spectral intensities are presumed to follow Gamma distribution. The conditional probability distribution of x_i given sublabel j and label c is presented as

$$p(x_i|Y_i = j, Z_i = c) = p_{icj}\left(x_i|\boldsymbol{\theta}_{cj}\right) = \frac{x_i^{\alpha_{cj}-1}}{\Gamma\left(\alpha_{cj}\right)\beta_{cj}^{\alpha_{cj}}}\exp\left(-\frac{x_i}{\beta_{cj}}\right) \quad (1)$$

where $p_{icj}\ (x_i|\boldsymbol{\theta}_{cj})$ presents Gamma distribution called the element; $\boldsymbol{\theta}_{cj} = \{\alpha_{cj}, \beta_{cj}\}$ is the set of parameters for Gamma element; α_{cj} is the shape parameter; β_{cj} is the scale parameter; and $\Gamma(\alpha_{cj}) = \int_0^\infty (a^{\alpha_{cj}-1}/e^a)da$ is Gamma function of α_{cj}. The prior probability of pixel i being assigned into subregion j of objective region c is presented as

$$p(Y_i = j|Z_i = c) = w_{icj} \quad (2)$$

where w_{icj} is called the element weight and satisfies the constraints $0 \le w_{icj} \le 1$ and $\sum_{j=1}^m w_{icj} = 1$.

The conditional probability distribution of x_i given label c can be expressed by combining (1) and (2) as

$$p(x_i|Z_i = c) = \sum_{Y_i} p(x_i|Y_i = j, Z_i = c)\,p(Y_i = j|Z_i = c). \quad (3)$$

Equation (3) can be further written as

$$p(x_i|Z_i = c) = p_{ic}(x_i|\boldsymbol{w}_{ic}, \boldsymbol{\theta}_c) = \sum_{j=1}^m w_{icj}\,p_{icj}\left(x_i|\boldsymbol{\theta}_{cj}\right) \quad (4)$$

where $p_{ic}\ (x_i|\boldsymbol{w}_{ic}, \boldsymbol{\theta}_c)$ is called the component of HGaMM; and $\boldsymbol{w}_{ic} = \{w_{icj};\ j = 1, 2, \ldots, m\}$ and $\boldsymbol{\theta}_c = \{\boldsymbol{\theta}_{cj};\ j = 1, 2, \ldots, m\}$ are the sets of element weights and parameters, respectively. Then, the prior probability of pixel i being assigned into the objective region c can be expressed as

$$p(Z_i = c) = \pi_{ic} \quad (5)$$

where π_{ic} is called the component weight, and it satisfies the constraints $0 \le \pi_{ic} \le 1$ and $\sum_{c=1}^k \pi_{ic} = 1$.

The marginal probability distribution of x_i can be expressed by combining (4) and (5) as

$$p(x_i) = \sum_{Z_i} p(Z_i = c)\,p(x_i|Z_i = c). \quad (6)$$

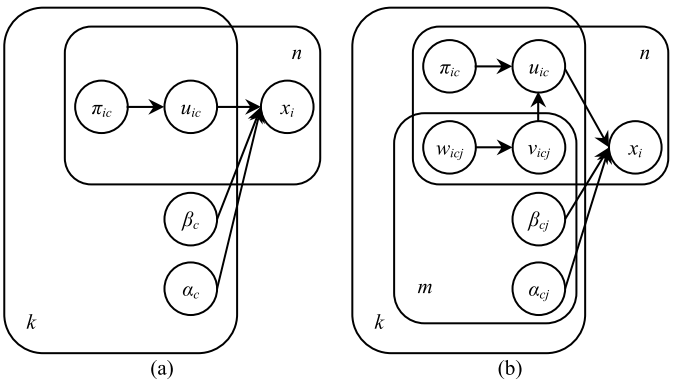


Fig. 1. Graphical models for GaMM and HGaMM. (a) GaMM. (b) HGaMM.

Equation (6) can be further written, given the set of model parameters $\boldsymbol{\Theta}_i$ for pixel i, as

$$p_i(x_i|\boldsymbol{\Theta}_i) = \sum_{c=1}^k \pi_{ic}\,p_{ic}(x_i|\boldsymbol{w}_{ic}, \boldsymbol{\theta}_c) = \sum_{c=1}^k \pi_{ic}\left(\sum_{j=1}^m w_{icj}\,p_{icj}\left(x_i|\boldsymbol{\theta}_{cj}\right)\right) \quad (7)$$

where $\boldsymbol{\Theta}_i = \{\boldsymbol{\pi}_i, \boldsymbol{w}_i, \boldsymbol{\theta}\}$, $\boldsymbol{\pi}_i = \{\pi_{ic};\ c = 1, 2, \ldots, k\}$, $\boldsymbol{w}_i = \{\boldsymbol{w}_{ic};\ c = 1, 2, \ldots, k\}$, and $\boldsymbol{\theta} = \{\boldsymbol{\theta}_c;\ c = 1, 2, \ldots, k\}$.

Following the Bayesian theory, given model parameters $\{\boldsymbol{\pi}^{(t)}, \boldsymbol{w}^{(t)}, \boldsymbol{\theta}^{(t)}\}$ of iteration t, the posterior probability u_{ic} of pixel i being assigned into the objective region c can be expressed as

$$u_{ic}^{(t)} = p(Z_i = c|x_i) = \frac{p(x_i|Z_i = c)\,p(Z_i = c)}{p(x_i)} = \frac{\pi_{ic}^{(t)}\sum_{j=1}^m w_{icj}^{(t)}\,p_{icj}\left(x_i\middle|\boldsymbol{\theta}_{cj}^{(t)}\right)}{\sum_{c'=1}^k \pi_{ic'}^{(t)}\sum_{j=1}^m w_{ic'j}^{(t)}\,p_{ic'j}\left(x_i\middle|\boldsymbol{\theta}_{c'j}^{(t)}\right)}. \quad (8)$$

Similarly, the element posterior probability, which presents the probability of pixel i being assigned into subregion j of objective region c, can be expressed as

$$v_{icj}^{(t)} = p(Y_i = j|x_i, Z_i = c) = \frac{p(x_i|Y_i = j, Z_i = c)\,p(Y_i = j|Z_i = c)}{p(x_i|Z_i = c)} = \frac{w_{icj}^{(t)}\,p_{icj}\left(x_i\middle|\boldsymbol{\theta}_{cj}^{(t)}\right)}{\sum_{j'=1}^m w_{icj'}^{(t)}\,p_{icj'}\left(x_i\middle|\boldsymbol{\theta}_{cj'}^{(t)}\right)}. \quad (9)$$

Equation (7) is the distribution of spectral intensity x_i modeled by HGaMM. When $m = 1$, the element weights are simplified as 1, and the components are simplified as Gamma distributions. Then, HGaMM is reduced as GaMM. The graphical model for GaMM is presented in Fig. 1(a). Hence, HGaMM can be viewed as generalized GaMM. The graphical model for HGaMM is shown in Fig. 1(b).

B. HGaMM-HMRF Segmentation Model

Given a high-resolution SAR image, the conditional probability distribution of x_i can be modeled by HGaMM in (7)

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and assumed that the pixels are independent of each other given the latent label variables. Then, the joint distribution of spectral intensities can be presented as

$$p(\boldsymbol{x}|\boldsymbol{\Theta}) = \prod_{i=1}^{n} p_i(x_i|\boldsymbol{\Theta}_i) = \prod_{i=1}^{n}\left[\sum_{c=1}^{k}\pi_{ic}\left(\sum_{j=1}^{m} w_{icj}\frac{x_i^{\alpha_{cj}-1}\exp\left(-x_i/\beta_{cj}\right)}{\Gamma\left(\alpha_{cj}\right)\beta_{cj}^{\alpha_{cj}}}\right)\right] \tag{10}$$

where $\boldsymbol{\Theta} = \{\boldsymbol{\Theta}_i;\ i = 1, 2, \ldots, n\}$, which can be written as $\boldsymbol{\Theta} = \{\boldsymbol{\pi}, \boldsymbol{w}, \boldsymbol{\alpha}, \boldsymbol{\beta}\}$, where $\boldsymbol{\pi} = \{\boldsymbol{\pi}_i;\ i = 1, 2, \ldots, n\}$, $\boldsymbol{w} = \{\boldsymbol{w}_i;\ i = 1, 2, \ldots, n\}$, $\boldsymbol{\alpha} = \{\alpha_{cj};\ c = 1, 2, \ldots, k,\ j = 1, 2, \ldots, m\}$, and $\boldsymbol{\beta} = \{\beta_{cj}; c = 1, 2, \ldots, k,\ j = 1, 2, \ldots, m\}$. Equation (10) is called the HGaMM-based statistical model.

To reduce the effect of image noise, the spatial relations of pixels can be incorporated into the prior probability by the MRF [15], [30]. The component weights can be defined with the posterior probabilities of neighboring pixels and can be presented as

$$\pi_{ic}^{(t+1)} = \frac{\exp\left(\frac{\eta}{\#\boldsymbol{N}_i}\sum_{i'\in\boldsymbol{N}_i} u_{i'c}^{(t)}\right)}{\sum_{c'=1}^{k}\left(\exp\left(\frac{\eta}{\#\boldsymbol{N}_i}\sum_{l\in N_i} u_{lc'}^{(t)}\right)\right)} \tag{11}$$

where η is a coefficient controlling the strength of smoothing noise; $\boldsymbol{N}_i$ presents the set of indexes of neighboring pixels; l is the index of neighboring pixels; and # presents the counting symbol. The size of the neighboring window is 3×3 pixels.

According to the structure of (10), the element weight and the scale parameter can be calculated using maximum likelihood estimation. Obtaining the optimum number of components and estimating the shape parameter in closed form can be difficult. Hence, the proposed algorithm assumes that the number of components satisfies the Poisson distribution with coefficient r, which can be written as

$$p(k) = \frac{r^k\exp(-r)}{k!}. \tag{12}$$

Due to the shape parameter contained in the Gamma function, its expression cannot be obtained. The prior distribution of the shape parameters can be defined by Gaussian distribution, denoted by $\alpha_{cj} \sim N(\mu_\alpha, \sigma_\alpha)$, and is independent in components and elements. Then, the prior distribution of the shape parameter is written as

$$p(\boldsymbol{\alpha}|k) = \prod_{c=1}^{k}\prod_{j=1}^{m} p\left(\alpha_{cj}|k\right) = \prod_{c=1}^{k}\prod_{j=1}^{m}\frac{1}{\sqrt{2\pi\sigma_\alpha^2}}\exp\left(-\frac{\left(\alpha_{cj}-\mu_\alpha\right)^2}{2\sigma_\alpha^2}\right). \tag{13}$$

The posterior distribution of model parameters can be constructed by the Bayesian theory as a segmentation model, which is written as

$$\begin{aligned} &p(k, \boldsymbol{\alpha}|\boldsymbol{x}, \boldsymbol{\pi}, \boldsymbol{w}, \boldsymbol{\beta}) \\ &\propto p(\boldsymbol{x}|k, \boldsymbol{\pi}, \boldsymbol{w}, \boldsymbol{\alpha}, \boldsymbol{\beta})p(\boldsymbol{\alpha}|k)p(k) \\ &= \prod_{i=1}^{n}\left[\sum_{c=1}^{k}\pi_{ic}\sum_{j=1}^{m} w_{icj}\frac{x_i^{\alpha_{cj}-1}\exp\left(-x_i/\beta_{cj}\right)}{\Gamma\left(\alpha_{cj}\right)\beta_{cj}^{\alpha_{cj}}}\right] \\ &\quad\times\prod_{c=1}^{k}\prod_{j=1}^{m}\left(2\pi\sigma_\alpha^2\right)^{-1/2}\exp\left(-\frac{\left(\alpha_{cj}-\mu_\alpha\right)^2}{2\sigma_\alpha^2}\right)\times\frac{r^k\exp(-r)}{k!}. \end{aligned} \tag{14}$$

C. Parameter Estimation

A novel EM/ADMCMC algorithm is proposed to maximize (14) in order to obtain accurate segmentation of high-resolution SAR images. The proposed algorithm can estimate the model parameters (especially the shape parameter) and determine the optimum number of components. The explicitly expressed parameters, w_{icj} and β_{cj}, can be calculated using maximum likelihood estimation, and the implicit parameters, k and α_{cj}, can be sampled given w_{icj} and β_{cj}. The specific solving process is as follows:

1) Estimation of Explicit Parameters: Given k and α_{cj}, the logarithm likelihood function with respect to w_{icj} and β_{cj} is written as

$$L(\boldsymbol{w}, \boldsymbol{\beta}) = \sum_{i=1}^{n}\log\left[\sum_{c=1}^{k}\pi_{ic}\sum_{j=1}^{m} w_{icj}\frac{x_i^{\alpha_{cj}-1}\exp\left(-x_i/\beta_{cj}\right)}{\Gamma\left(\alpha_{cj}\right)\beta_{cj}^{\alpha_{cj}}}\right]. \tag{15}$$

The posterior probabilities of the components satisfy its constraint. Then, $\log(\sum_{c=1}^{k} u_{ic}G_c) \geq \sum_{c=1}^{k} u_{ic}\log(G_c)$ can be obtained based on the Jensen inequality. Applying this equation to (15), the inequality of logarithm likelihood is presented as

$$L(\boldsymbol{w}, \boldsymbol{\beta}) \geq \sum_{i=1}^{n}\sum_{c=1}^{k} u_{ic}^{(t)} \times\left\{\log\pi_{ic}^{(t)} + \log\left[\sum_{j=1}^{m} w_{icj}\,p_{icj}\left(x_i\middle|\alpha_{cj}^{(t)}, \beta_{cj}\right)\right]\right\}. \tag{16}$$

Similarly, the posterior probabilities of the elements satisfy the constraint $\sum_{j=1}^{m} v_{icj} = 1$. Based on the Jensen inequality, $\log(\sum_{j=1}^{m} v_{icj}H_j) \geq \sum_{j=1}^{m} v_{icj}\log(H_j)$ can be obtained, and (16) can be further presented as

$$L(\boldsymbol{w}, \boldsymbol{\beta}) \geq \sum_{i=1}^{n}\sum_{c=1}^{k} u_{ic}^{(t)}\left\{\log\pi_{ic}^{(t)} + \sum_{j=1}^{m} v_{icj}^{(t)} \times\left[\log w_{icj} + \log p_{icj}\left(x_i|\alpha_{cj}^{(t)}, \beta_{cj}\right)\right]\right\}. \tag{17}$$

The right term of (17) presents the lower bound of $L(\boldsymbol{w}, \boldsymbol{\beta})$. Maximizing $L(\boldsymbol{w}, \boldsymbol{\beta})$ can be converted to maximize its lower

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bound. Hence, the bound can be denoted as the new objective function

$$J(\boldsymbol{w},\boldsymbol{\beta}) = \sum_{i=1}^{n}\sum_{c=1}^{k} u_{ic}^{(t)} \left\{ \log \pi_{ic}^{(t)} + \sum_{j=1}^{m} v_{icj}^{(t)} \times \left[\log w_{icj} + \left(\alpha_{cj}^{(t)} - 1\right)\log x_i - \log\Gamma\left(\alpha_{cj}^{(t)}\right) - \log \beta_{cj}^{\alpha_{cj}^{(t)}} - \frac{x_i}{\beta_{cj}} \right] \right\}. \quad (18)$$

According to (18), the derivation of $J(\boldsymbol{w}, \boldsymbol{\beta})$ with respect to β_{cj} can be derived and set to 0. Then, the expression of β_{cj} at $t+1$ iteration can be calculated as

$$\beta_{cj}^{(t+1)} = \frac{\sum_{i=1}^{n} u_{ic}^{(t)} v_{icj}^{(t)} x_i}{\sum_{i=1}^{n} u_{ic}^{(t)} v_{icj}^{(t)} \alpha_{cj}^{(t)}}. \quad (19)$$

Lagrange multiplier method is used to estimate w_{icj} for satisfying its constraints. The Lagrange function with respect to w_{icj} can be presented as

$$J_w = J(\boldsymbol{w},\boldsymbol{\beta}) - \sum_{i=1}^{n}\sum_{c=1}^{k} \gamma_{ic}\left(\sum_{j=1}^{m} w_{icj} - 1\right) \quad (20)$$

where γ_{ic} is the Lagrange coefficient. The derivations of J_w with respect to w_{icj} and γ_{ic} can be derived and are set to be 0. The expression of w_{icj} at $t+1$ iteration can be calculated as

$$w_{icj}^{(t+1)} = \frac{v_{icj}^{(t)}}{\sum_{j'=1}^{m} v_{icj'}^{(t)}}. \quad (21)$$

2) Implicit Parameter Estimation: The ADMCMC algorithm is designed to simulate (14) given $\boldsymbol{w}$, $\boldsymbol{\beta}$, and $\boldsymbol{\pi}$. Updating the shape parameter and adding or deleting component operations are designed to optimize $\boldsymbol{\alpha}$ and k, respectively. Let the set of current parameters be $\boldsymbol{\Theta}^{(t)} = \{\boldsymbol{\pi}^{(t)}, \boldsymbol{w}^{(t)}, \boldsymbol{\alpha}^{(t)}, \boldsymbol{\beta}^{(t)}\}$, and the number of current components be $k^{(t)}$. Let the set of candidate parameters be $\boldsymbol{\Theta}^* = \{\boldsymbol{\pi}^*, \boldsymbol{w}^*, \boldsymbol{\alpha}^*, \boldsymbol{\beta}^*\}$, and the number of candidate components be k^*.

In updating the shape parameter, α_{cj} is randomly selected from $\boldsymbol{\alpha}^{(t)}$, where $c \in \{1, 2, \ldots, k\}$ and $j \in \{1, 2, \ldots, m\}$. The candidate shape parameter α_{cj}^* is randomly selected from its prior distribution $N(\alpha_{cj}, 0.5)$. And the variance is an empirical value, which is too large to miss the optimal shape parameter and too small to cause low efficiency. Then, the candidate shape parameters can be presented as $\boldsymbol{\alpha}^* = \{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_c^*, \ldots, \boldsymbol{\alpha}_k\}$, where $\boldsymbol{\alpha}_c^* = \{\alpha_{c1}, \ldots, \alpha_{cj}^*, \ldots, \alpha_{cm}\}$. The acceptance rate of $\boldsymbol{\alpha}^*$ can be calculated as

$$a\left(\boldsymbol{\alpha}^{(t)}, \boldsymbol{\alpha}^*\right) = \min\left\{1, \frac{\exp\left(-\frac{1}{2\sigma_\alpha^2}\left(\alpha_{cj}^* - \mu_\alpha\right)^2\right)}{\exp\left(-\frac{1}{2\sigma_\alpha^2}\left(\alpha_{cj} - \mu_\alpha\right)^2\right)} \times \frac{\prod_{i=1}^{n}\left[\sum_{c=1}^{k} \pi_{ic}^{(t)} \sum_{j=1}^{m} w_{icj}^{(t)} p_{icj}\left(x_i \middle| \alpha_{cj}^*, \beta_{cj}^{(t)}\right)\right]}{\prod_{i=1}^{n}\left[\sum_{c=1}^{k} \pi_{ic}^{(t)} \sum_{j=1}^{m} w_{icj}^{(t)} p_{icj}\left(x_i \middle| \alpha_{cj}, \beta_{cj}^{(t)}\right)\right]}\right\}. \quad (22)$$

TABLE I
SETTING CONSTANTS OF THE PROPOSED ALGORITHM

T	η	r	μ_α	σ_α	m
300	0.7	4	16	8	2

If $a(\boldsymbol{\alpha}^{(t)}, \boldsymbol{\alpha}^*) = 1$, $\boldsymbol{\alpha}^*$ can be accepted and $\boldsymbol{\alpha}^{(t+1)}$ is equal to $\boldsymbol{\alpha}^*$; otherwise, $\boldsymbol{\alpha}^{(t+1)}$ is equal to $\boldsymbol{\alpha}^{(t)}$.

In the component addition or deletion procedure, one of the two operations is randomly implemented in the iterations. In the component addition operation, let the candidate component number be $k^* = k + 1$. The components $\boldsymbol{\pi}_{k+1}$, $\boldsymbol{w}_{k+1}$, and $\boldsymbol{\theta}_{k+1}$ are then added as follows: $\boldsymbol{\pi}_{k+1} = \{\pi_{i(k+1)};\ i = 1, 2, \ldots, n\}$ is randomly generated from (0, 1). The set $\{\boldsymbol{\pi}_1, \ldots, \boldsymbol{\pi}_k, \boldsymbol{\pi}_{k+1}\}$ would be normalized to meet the constraint. Then, the satisfied weights can be presented as $\boldsymbol{\pi}^* = \{\boldsymbol{\pi}_1/(1+\boldsymbol{\pi}_{k^*}), \ldots, \boldsymbol{\pi}_k/(1+\boldsymbol{\pi}_{k^*}), \boldsymbol{\pi}_{k^*}/(1+\boldsymbol{\pi}_{k^*})\}$, as the set of candidate component weights. $\boldsymbol{w}_{k+1} = \{w_{i(k+1)j};\ i = 1, 2, \ldots, n,\ j = 1, 2, \ldots, m\}$ is randomly generated from (0, 1) and satisfies the constraints. Its candidate set is presented as $\boldsymbol{w}^* = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k, \boldsymbol{w}_{k+1}\}$. $\boldsymbol{\theta}_{k+1} = \{\boldsymbol{\theta}_{(k+1)j};\ j = 1, 2, \ldots, m\}$ is randomly generated from its prior Gaussian distribution, where $\boldsymbol{\theta}_{(k+1)j} = \{\alpha_{(k+1)j}, \boldsymbol{\beta}_{(k+1)j}\}$. Then, $\boldsymbol{\theta}^* = \{\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_k, \boldsymbol{\theta}_{k+1}\}$ is viewed as the candidate parameters of the Gamma element. The acceptance rate is presented as $a_a(\{k, \boldsymbol{\Theta}\}, \{k^*, \boldsymbol{\Theta}^*\}) = \min\{1, R\}$, where

$$R = \frac{\prod_{i=1}^{n}\left[\sum_{c=1}^{k^*} \pi_{ic}^* \sum_{j=1}^{m} w_{icj}^* p_{icj}\left(x_i \middle| \boldsymbol{\theta}_{cj}^*\right)\right]}{\prod_{i=1}^{n}\left[\sum_{c=1}^{k} \pi_{ic} \sum_{j=1}^{m} w_{icj} p_{icj}\left(x_i \middle| \boldsymbol{\theta}_{cj}\right)\right]} \times \frac{r}{k^*}. \quad (23)$$

If $k > 2$, the component deletion operation can be carried out. The candidate component number is $k^* = k - 1$. The index c is randomly selected from $\{1, 2, \ldots, k\}$. Then, $\boldsymbol{\pi}_c$, $\boldsymbol{w}_c$, and $\boldsymbol{\theta}_c$ are correspondingly deleted. The set $\{\boldsymbol{\pi}_1, \ldots, \boldsymbol{\pi}_{c-1}, \boldsymbol{\pi}_{c+1}, \ldots, \boldsymbol{\pi}_k\}$ would be normalized to meet the constraint, and can be further written as $\boldsymbol{\pi}^* = \{\boldsymbol{\pi}_1/(1-\boldsymbol{\pi}_c), \ldots, \boldsymbol{\pi}_{c-1}/(1-\boldsymbol{\pi}_c), \boldsymbol{\pi}_{c+1}/(1-\boldsymbol{\pi}_c), \ldots, \boldsymbol{\pi}_k(1-\boldsymbol{\pi}_c)\}$, as the candidate weight. Moreover, $\boldsymbol{w}^* = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{c-1}, \boldsymbol{w}_{c+1}, \ldots, \boldsymbol{w}_k\}$ is the set of candidate weights, and $\boldsymbol{\theta}^* = \{\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{c-1}, \boldsymbol{\theta}_{c+1}, \ldots, \boldsymbol{\theta}_k\}$ is the set of parameters. The acceptance rate is calculated as $a_d(\{k, \boldsymbol{\Theta}\}, \{k^*, \boldsymbol{\Theta}^*\}) = \min\{1, 1/R\}$, due to it being the antithesis of the component addition operation.

If $a_a(\{k, \boldsymbol{\Theta}\}, \{k^*, \boldsymbol{\Theta}^*\}) = 1$ or $a_d(\{k, \boldsymbol{\Theta}\}, \{k^*, \boldsymbol{\Theta}^*\}) = 1$, then $k^{(t+1)}$ is equal to k^* and $\boldsymbol{\Theta}^{(t+1)}$ is equal to $\boldsymbol{\Theta}^*$; otherwise, $k^{(t+1)}$ is equal to $k^{(t)}$ and $\boldsymbol{\Theta}^{(t+1)}$ is equal to $\boldsymbol{\Theta}^{(t)}$. The result of segmenting SAR image is represented by the set of labels $\boldsymbol{z} = \{z_i;\ i = 1, 2, \ldots, n\}$. The label of pixel i is obtained by maximizing the posterior probability u_{ic}. The parameter estimation of the proposed HGaMM algorithm is summarized in Fig. 2.

III. EXPERIMENTAL RESULTS AND DISCUSSION

Numerous experiments were implemented using FCM-HMRF [48], Gamma [13], GMM [23], SMM [30], GaMM, and the proposed algorithm on simulated and real high-resolution SAR images, under the experimental

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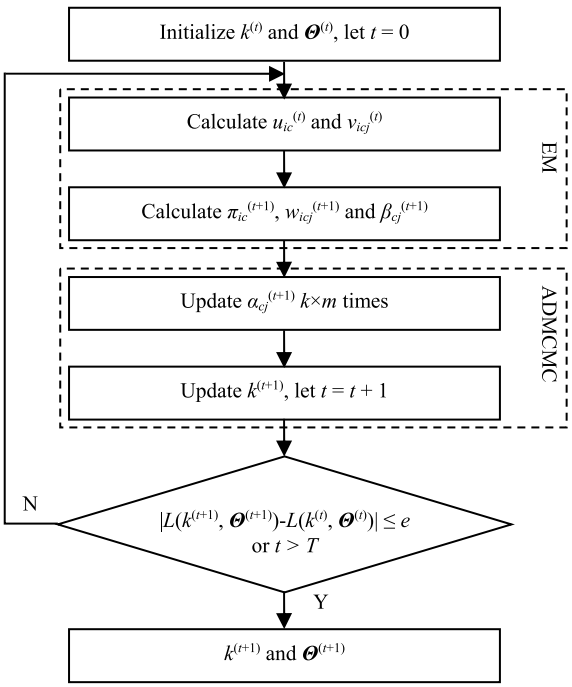


Fig. 2. Parameter estimation of the proposed algorithm.

TABLE II
PARAMETERS OF GAMMA DISTRIBUTION FOR GENERATING FIG. 3(b)

Parameters		α_{cj}	β_{cj}	w_{cj}
Region 1	Distribution 1 ($j = 1$)	4	2	0.6
($c = 1$)	Distribution 2 ($j = 2$)	12	3	0.4
Region 2	Distribution 1 ($j = 1$)	25	3	0.6
($c = 2$)	Distribution 2 ($j = 2$)	17	8	0.4
Region 3	Distribution 1 ($j = 1$)	30	6	0.6
($c = 3$)	Distribution 2 ($j = 2$)	30	7	0.4

environment of MATLAB R2016a software. Table I lists the setting constants in this article. T is the maximum iteration. The larger the value of $\eta \in (0, 1)$, the stronger the ability for smoothing noise. However, if the value becomes extremely large, the procedure would cause over-segmentation. Hence, η was set to be 0.7, according to the numerous experiments. The parameter r was set by the approximate average of the number of objective regions, while μ_α and σ_α were based on prior knowledge [13]. Parameter m was set to be 2, enough to build the complex statistical model of spectral intensities in SAR images. If m is extremely large, the huge estimated parameters would complicate the segmentation model.

A. Experiment of Simulated SAR Image

Two simulated images were used for segmentation experiments. The boundaries between regions are curved in the first image and are horizontal and vertical in the second image. They can verify the segmentation performance of the proposed algorithm for different shape regions.

In the first experiment, a standard image with three homogeneous regions (labeled as 1–3) was used and shown in Fig. 3(a). The simulated SAR image can be then generated, as shown in Fig. 3(b). The spectral intensity for each region was randomly generated by the weighted sum of two Gamma distributions, and the parameters are listed in Table II. Fig. 3(c)–(i) presents the segmentation results utilizing FCM-HMRF, Gamma, GMM, SMM, GaMM, and the proposed algorithm ($m = 2$ and 3). Although traditional algorithms consider the pixel's local relation, many pixels were segmented into the wrong regions. For example, white pixels in region 2 of Fig. 3(c) were wrongly segmented by FCM-HMRF algorithm; the result in Fig. 3(d) was slightly better than Fig. 3(c), but some pixels were still wrongly segmented in region 2; many pixels were erroneously segmented in region 2, and a few pixels were wrongly segmented in regions 1 and 3 of Fig. 3(e); a number of pixels were wrongly segmented by SMM and GaMM algorithm in Fig. 3(f) and (g). In Fig. 3(h) and (i), each region can be segmented accurately when $m = 2$ and 3, having few pixels erroneously segmented. And the proposed algorithm can obtain better results whether m is set to be 2 or 3. This suggests that the proposed HGaMM algorithm yielded the best results compared with the other segmentation algorithms.

To evaluate the proposed algorithm's modeling ability, the results were presented in fitting histograms of the first simulated SAR image. FCM-HMRF algorithm is a clustering algorithm, so it is not used for fitting histogram. Therefore, Gamma, GMM, SMM, GaMM, and proposed algorithms are used to fit histograms. As shown in Fig. 4, the horizontal and vertical axes are the spectral intensity and its counts, respectively. The blue zones are the histograms for each region. The curves are the fitting results using the various segmentation algorithms presented in different colors. The histogram is asymmetric and peaky for region 1, heavy-tailed and bimodal for region 2, and flat for region 3. The results show that the traditional algorithms were unable to fit complex histograms. For example, the Gamma and GaMM algorithms cannot fit the peak and bimodal histograms of regions 1 and 2. The GMM and SMM algorithms failed to fit the asymmetric histograms of regions 1 and 2, while the SMM algorithm cannot accurately fit the flat histogram of region 3. The inaccurate fitting results of the traditional algorithms resulted in the erroneous segmentation of pixels, while the proposed algorithm was able to accurately fit the peak, bimodal, asymmetric, heavy-tailed, and flat histograms using weighted Gamma elements as components. This indicates that the proposed algorithm is capable of modeling complex distributions of spectral intensities and can obtain better segmentation results.

To quantitatively evaluate the results in Fig. 3, Table III lists the segmentation accuracy results in the first experiment. The accuracy can be calculated through the confusion matrix, including production accuracy (PA), user's accuracy (UA), overall accuracy (OA), and kappa coefficient (KC). When the accuracies generated by each algorithm were compared, the results showed that region 2 produced the lowest accuracy in each algorithm, having many pixels segmented erroneously in this region. The lowest PA and UA produced by the different algorithms were as follows: FCM-HMRF (81% and 78%), Gamma (96% and 90%), GMM (61% and 73%), SMM (96%

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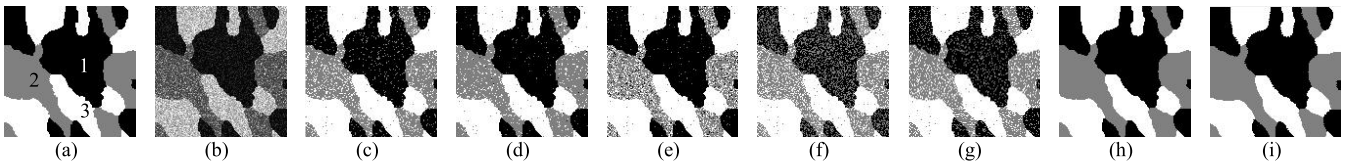


Fig. 3. First experiment. (a) Standard image. (b) Simulated SAR image. (c) FCM-HMRF algorithm. (d) Gamma algorithm. (e) GMM algorithm. (f) SMM algorithm. (g) GaMM algorithm. (h) Proposed algorithm ($m = 2$). (i) Proposed algorithm ($m = 3$).

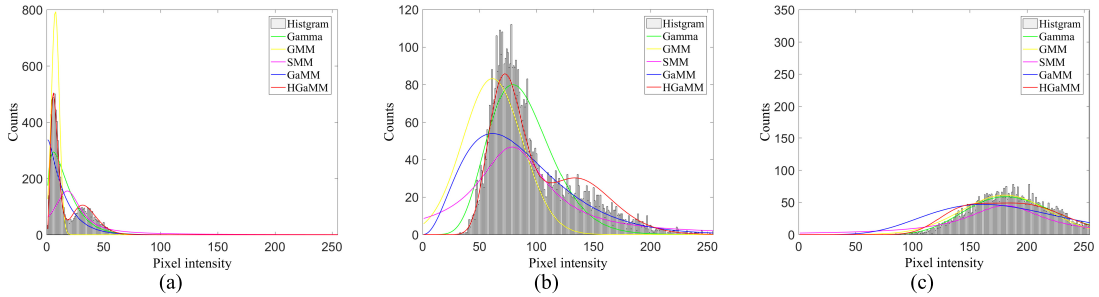
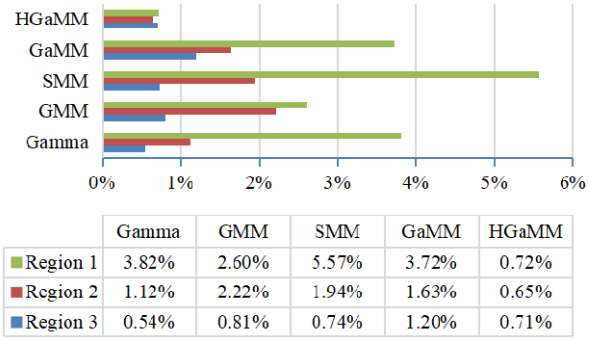


Fig. 4. Results of fitting histograms of the first simulated SAR image. (a)–(c) Regions 1–3.

TABLE III
ACCURACY EVALUATIONS OF SEGMENTATION RESULTS FOR THE FIRST SIMULATED SAR IMAGE

Algorithm	Accuracy	First simulated SAR image		
		Region 1	Region 2	Region 3
FCM-HMRF algorithm	PA (%)	96.43	84.99	81.86
	UA (%)	88.75	78.36	95.45
	OA (%)		88.23	
	KC		0.82	
Gamma algorithm	PA (%)	99.59	95.69	92.03
	UA (%)	98.06	90.27	98.52
	OA (%)		95.88	
	KC		0.93	
GMM algorithm	PA (%)	100	61.27	80.02
	UA (%)	64.53	73.38	98.81
	OA (%)		78.21	
	KC		0.70	
SMM algorithm	PA (%)	95.50	96.13	79.91
	UA (%)	98.50	67.16	99.27
	OA (%)		89.39	
	KC		0.84	
GaMM algorithm	PA (%)	100	78.23	83.65
	UA (%)	83.57	79.13	98.94
	OA (%)		87.20	
	KC		0.81	
The proposed algorithm ($m = 2$)	PA (%)	99.55	99.17	99.62
	UA (%)	99.77	99.30	99.25
	OA (%)		99.46	
	KC		0.99	
The proposed algorithm ($m = 3$)	PA (%)	97.57	97.05	95.41
	UA (%)	96.96	96.01	97.05
	OA (%)		96.71	
	KC		0.95	

and 67%), and GaMM (78% and 79%). For the OAs and KCs of traditional algorithms, only Gamma algorithm was higher than 90% and 0.90. In comparison, the proposed algorithm for the lowest PAs and UAs yielded 99% and 99% when $m = 2$, and 95% and 96% when $m = 3$, which were both higher than



	Gamma	GMM	SMM	GaMM	HGaMM
Region 1	3.82%	2.60%	5.57%	3.72%	0.72%
Region 2	1.12%	2.22%	1.94%	1.63%	0.65%
Region 3	0.54%	0.81%	0.74%	1.20%	0.71%

Fig. 5. Fitting errors of histograms for the first simulated SAR image.

other algorithms. Its OAs and KCs were also higher than any of the other algorithms. Hence, these results suggest that the proposed algorithm can obtain more accurate results whether m is set to be 2 or 3.

To quantitatively evaluate the results in Fig. 4, the fitting errors were calculated by comparing the real histogram values with estimated values. The smaller the fitting error, the higher the fitting accuracy. The fitting error can be expressed as $E_f = (\sum_{h=1}^{256} (p_h - p'_h)^2)^{1/2} \times 100\%$, where p_h and p'_h are the real and estimated probability value, respectively, and h is the index of spectral intensity. The calculated fitting errors were presented in Fig. 5. The errors of the proposed algorithm were lower than 0.72%. For the traditional algorithms, the resulting model histograms yielded large fitting errors, particularly in region 1. Although the error of the Gamma algorithm was lower for region 3 than the proposed algorithm, its errors for regions 2 and 3 were comparatively larger. These quantitative results suggest that the proposed algorithm can fit complex histograms with high accuracy and build accurate statistical models of spectral intensities.

In the second experiment, a standard image with four homogeneous regions (labeled as 1–4) was used and is shown

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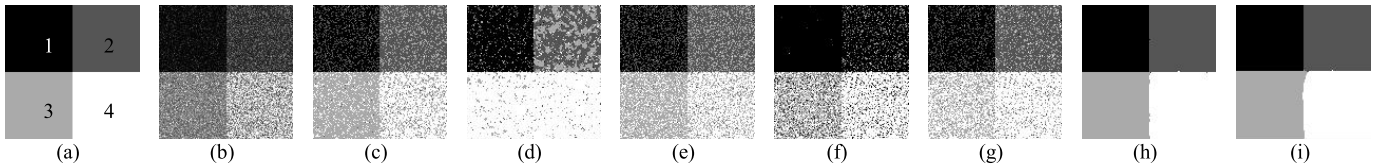


Fig. 6. Second experiment. (a) Standard image. (b) Simulated SAR image. (c) FCM-HMRF algorithm. (d) Gamma algorithm. (e) GMM algorithm. (f) SMM algorithm. (g) GaMM algorithm. (h) Proposed algorithm ($m = 2$). (i) Proposed algorithm ($m = 3$).

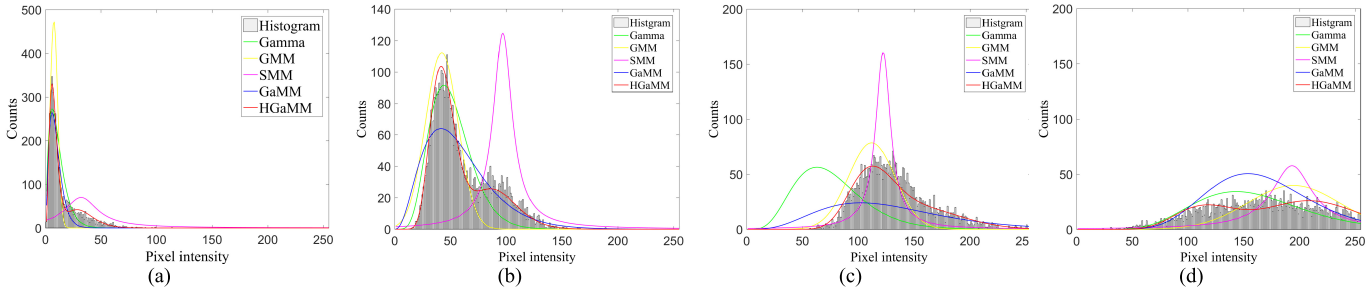
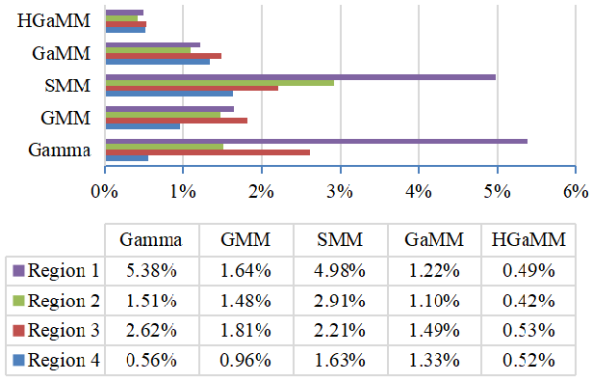


Fig. 7. Results of fitting histograms of the second simulated SAR image. (a)–(d) Regions 1–4.

TABLE IV
PARAMETERS OF GAMMA DISTRIBUTION FOR GENERATING FIG. 6(b)

Parameters		α_{cj}	β_{cj}	w_{cj}
Region 1	Distribution 1 ($j = 1$)	4	2	0.7
($c = 1$)	Distribution 2 ($j = 2$)	4	8	0.3
Region 2	Distribution 1 ($j = 1$)	15	3	0.7
($c = 2$)	Distribution 2 ($j = 2$)	23	4	0.3
Region 3	Distribution 1 ($j = 1$)	30	4	0.7
($c = 3$)	Distribution 2 ($j = 2$)	20	8	0.3
Region 4	Distribution 1 ($j = 1$)	6	30	0.7
($c = 4$)	Distribution 2 ($j = 2$)	10	21	0.3



	Gamma	GMM	SMM	GaMM	HGaMM
Region 1	5.38%	1.64%	4.98%	1.22%	0.49%
Region 2	1.51%	1.48%	2.91%	1.10%	0.42%
Region 3	2.62%	1.81%	2.21%	1.49%	0.53%
Region 4	0.56%	0.96%	1.63%	1.33%	0.52%

Fig. 8. Fitting errors of histograms for the second simulated SAR image.

in Fig. 6(a). Fig. 6(b) shows the second simulated SAR image, where the spectral intensities are generated by the weighted sum of two Gamma distributions (the parameters are listed in Table IV). Fig. 6(c)–(i) displays the segmentation results using the traditional and the proposed algorithms. In Fig. 6(c), many pixels were wrongly segmented for each region although FCM-HMRF algorithm taken the pixels' spatial relation in account. In Fig. 6(d), the Gamma algorithm was unable to segment regions 3 and 4. In Fig. 6(e)–(g), while the local spatial relations of pixels had been considered, the SMM, GMM, and GaMM algorithms still yielded poor results, having many pixels segmented erroneously. In Fig. 6(h) and (i), the proposed algorithm was able to properly segment the four regions, with very few pixels wrongly segmented, whether m is set to be 2 or 3. This indicates that compared with traditional algorithms, the proposed algorithm is able to provide better results.

Fig. 7(a)–(d) shows the fitting histogram results for the second simulated SAR image. The histograms show peaky, heavy-tailed, asymmetric, bimodal, and flat features. The GMM (yellow) and SMM (pink) algorithms failed to fit the complex histograms, given their symmetric and unimodal components. The GaMM (black) algorithm showed slightly better fitting results but failed to accurately fit the bimodal histogram in Fig. 7(b) and heavy-tailed histograms in Fig. 7(c) and (d). In comparison, the proposed (red) algorithm with weighted elements was able to fit the complex histograms more accurately. This indicates that the proposed algorithm performs better in modeling complex distribution of spectral intensities than traditional segmentation algorithms.

Table V lists the resulting segmentation accuracies from the quantitative evaluation. The results show the traditional segmentation algorithms yielded markedly low accuracies, especially for regions 2 and 3, and their OAs and KCs were lower than 77% and 0.70%. In comparison, the proposed algorithm showed better results, with accuracies of each region above 99% and 96% when m was set to be 2 and 3. Its OAs and KCs were higher than 98% and 0.98%, which were both higher than traditional algorithms.

Fig. 8 shows the curve fitting errors for the histograms in Fig. 7. The proposed algorithm obtained the lowest errors for each histogram, which ranged between 0.42% and 0.53%. Its errors were lower than traditional algorithms, especially region 1 using the Gamma and SMM algorithms. Hence, the proposed algorithm performs better in modeling complex distributions.

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TABLE V
ACCURACY EVALUATIONS OF SEGMENTATION RESULTS FOR THE SECOND SIMULATED SAR IMAGE

Algorithm	Accuracy	First simulated SAR image			
		Region 1	Region 2	Region 3	Region 4
FCM-HMRF algorithm	PA (%)	74.93	76.76	83.30	70.53
	UA (%)	100.00	73.07	62.47	81.38
	OA (%)	76.38			
	KC	0.70			
Gamma algorithm	PA (%)	99.26	85.11	14.95	50.07
	UA (%)	88.01	60.01	7.40	95.80
	OA (%)	62.81			
	KC	0.55			
GMM algorithm	PA (%)	99.97	68.27	54.42	70.35
	UA (%)	70.17	70.14	70.34	68.82
	OA (%)	69.87			
	KC	0.63			
SMM algorithm	PA (%)	53.30	35.60	66.69	67.49
	UA (%)	99.83	13.31	44.04	73.75
	OA (%)	57.73			
	KC	0.51			
GaMM algorithm	PA (%)	99.63	74.87	55.95	59.09
	UA (%)	78.76	78.47	43.51	81.76
	OA (%)	70.62			
	KC	0.70			
The proposed algorithm ($m = 2$)	PA (%)	99.98	99.95	99.59	99.95
	UA (%)	100	99.93	100	99.54
	OA (%)	99.87			
	KC	0.99			
The proposed algorithm ($m = 3$)	PA (%)	100.00	100.00	97.65	96.88
	UA (%)	98.44	98.44	98.27	99.29
	OA (%)	98.61			
	KC	0.98			

TABLE VI
PARAMETER ESTIMATION FOR TWO SIMULATED SAR IMAGES

Images	Para-meters	Real / Estimated parameters			
		Region 1	Region 2	Region 3	Region 4
The first image	α	4/4.24	25/24.21	30/32.59	-
		12/11.94	17/18.47	30/28.09	-
	β	2/1.82	3/3.08	6/5.90	-
		3/3.23	8/7.84	7/7.07	-
The second image	α	4/4.19	15/15.30	20/20.37	6/6.36
		4/4.21	23/22.43	30/29.67	10/9.54
	β	2/1.88	3/2.90	4/3.88	30/27.57
		8/7.67	4/4.30	8/7.80	21/23.14

Table VI lists the resulting parameter estimation for two simulated images. It shows that the estimated errors were less than 2.59. For the first simulated image, its errors of α for region 3 were relatively large (2.59 and 1.81). For the second simulated image, its errors of β for region 4 were 2.43 and 2.14, which were higher than other regions. The reason for the large error was wide spectral range and strong heterogeneity of spectral intensities in the corresponding region. These results suggest that the proposed algorithm can obtain accurate results of parameter estimation.

Fig. 9 shows the point diagrams displaying the change in component number for every iteration. Fig. 9(a) shows the results for the first simulated SAR image, while Fig. 9(b) shows for the second simulated SAR image. The horizontal and vertical axes represent the numbers of iterations and components, respectively. The initial component number was set to be 2. After about 50–60 iterations, the component number for the two simulated images stabilized at 3 in Fig. 9(a) and 4 in Fig. 9(b). These final component numbers correctly matched the numbers of homogeneous regions in the standard images. This suggests that the proposed algorithm can obtain the optimum number of components.

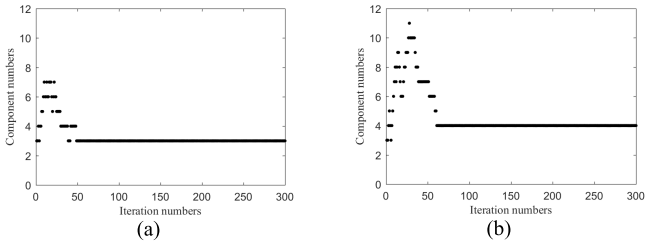


Fig. 9. Point diagrams of the component numbers changing with iterations. (a) and (b) First and the second simulated SAR images.

TABLE VII
SEGMENTATION TIME OF RESULTS FOR TWO SIMULATED SAR IMAGES

Algorithms	Time (s)	
	The first image	The second image
FCM-HMRF algorithm	28.26	24.78
Gamma algorithm	727.80	725.54
GMM algorithm	31.82	33.13
SMM algorithm	169.74	253.06
GaMM algorithm	764.28	854.74
The proposed algorithm ($m = 2$)	67.93	66.73
The proposed algorithm ($m = 3$)	89.54	90.31

The time-consuming of segmentation algorithm is affected by the computing power of device, programming language, and many other aspects. Without considering these external effects, the complexity of the proposed algorithm can be analyzed as follows. $O(kn_u)$ is the complexity of posterior probability for region, $O(kmn_v)$ is the complexity of posterior probability for subregion, $O(kn_\pi)$ is the complexity of component weight, $O(kmn_w)$ is the complexity of element weight, and the complexities of the shape and scale parameters are both $O(km)$. The complexities are related to the number of iterations T. And the complexity of the proposed algorithm is $T \times O[km\ (2 + n_v + n_w) + k\ (n_u + n_\pi)]$. Hence, its complexity is related to the scale of image and the numbers of components and elements.

Table VII lists the segmentation time of results for two simulated images. FCM-HMRF algorithm had the highest efficiency, because the algorithm is simple. The average time of GMM algorithm was about 32.47 s which was slightly more than FCM-HMRF algorithm, due to its parameters estimation used EM algorithm. The average times of the proposed algorithm were about 67.33 and 89.92 s when $m = 2$ and 3. Its efficiencies were better than Gamma, SMM, and GaMM algorithms, not as good as GMM algorithm. But its accuracies were higher than other algorithms including GMM algorithm. Moreover, the proposed algorithm can obtain the good results

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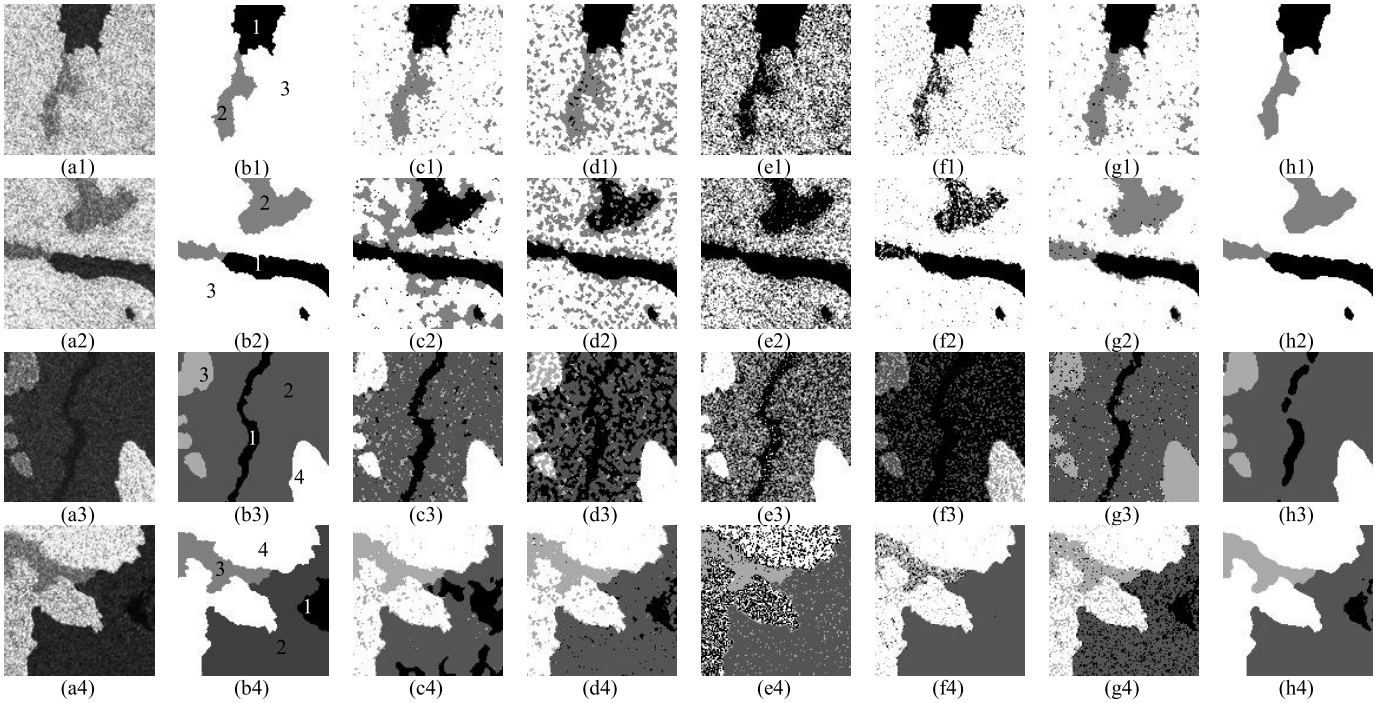


Fig. 10. Third experiment. (a1)–(a4) High-resolution SAR images. (b1)–(b4) Standard images. (c1)–(c4) FCM-HMRF algorithm. (d1)–(d4) Gamma algorithm. (e1)–(e4) GMM algorithm. (f1)–(f4) SMM algorithm. (g1)–(g4) GaMM algorithm. (h1)–(h4) Proposed algorithm.

when m was set to be 2 or 3, but the efficiency when $m = 2$ was better than that when $m = 3$. Hence, the proposed algorithm can present better performance when m is set to be 2.

B. Experiment of High-Resolution SAR Image

In the third experiment, high-resolution SAR images were segmented using traditional and proposed algorithms. As shown in Fig. 10(a1)–(a4), four SAR images were used which are 128 × 128 pixels from Radarsat-1 with 10-m spatial resolution, where the main ground object includes sea ice with different degrees of melting. Using visual discrimination of Fig. 10(a1)–(a4), standard images can be drawn, as shown in Fig. 10(b1)–(b4), where labels 1–4 represent different regions. In Fig. 10(c1)–(c4), several regions were wrongly segmented, in which the spectral intensities of pixels are similar. For example, regions 1 and 2 in Fig. 10(c2) and regions 3 and 4 in Fig. 10(c3) cannot be segmented. In Fig. 10(d1)–(d4), results of the Gamma algorithm were not better than the FCM-HMRF algorithm, and many pixel blocks with arbitrary shapes were segmented erroneously. Also, the black and gray regions were not be segmented in Fig. 10(d2). Although the GMM and SMM algorithms account for the pixels' local relation, many pixels still were wrongly segmented, as shown in Fig. 10(e1)–(e4) and (f1)–(f4). The two algorithms failed to segment the gray regions in Fig. 10(e1) and (e2) and (f1) and (f2) and the black region in Fig. 10(e4) and (f4). In Fig. 10(g1)–(g4), the GaMM algorithm obtained slightly better results, but wrongly segmented pixels and regions can still be found. For the proposed algorithm, fewer pixels were wrongly segmented, resulting in better segmentation of regions, as shown in Fig. 10(h1)–(h4). Visually, the resulting images suggest that the proposed algorithm performs better in segmenting high-resolution SAR images.

In the fourth experiment, two SAR images are segmented using traditional and proposed algorithms. Fig. 11(a1) and (a2) shows 512 × 512 pixels SAR images from Radarsat-1 with 10-m spatial resolution, where the main ground objects include architecture, farmland, and river. Using visual discrimination of Fig. 11(a1) and (a2), standard images can be drawn, as shown in Fig. 11(b1) and (b2), where labels 1–3 represent different regions. In Fig. 11(c1)–(g1) and (c2)–(g2), the traditional algorithms can segment the river (region 1), due to the high similarity of spectral intensities in the region. But lots of pixels were wrongly segmented for other regions, especially farmland (region 2), such as Fig. 11(c1) and (f2). In Fig. 11(h1) and (h2), the proposed algorithm can segment the regions more accurately, and fewer pixels were wrongly segmented. Visually, the resulting images suggest that the proposed algorithm has a better performance for segmenting large-scale SAR images.

Fig. 12 shows the OAs of segmentation results for Figs. 10 and 11 that quantitatively evaluates the performance of the proposed algorithm. According to the bar graph, the accuracies of the GMM algorithm were the lowest except for Figs. 10(a3) and 11(a1) and (a2). The lowest accuracy for Figs. 10(a3) and 11(a1)–(a2) was obtained using the SMM, FCM-HMRF, and SMM algorithms. Moreover, the accuracies of comparing algorithms were lower than 70% for Fig. 11(a1). While the highest accuracies were reached by the proposed algorithm, the accuracies were above 93% for each result.

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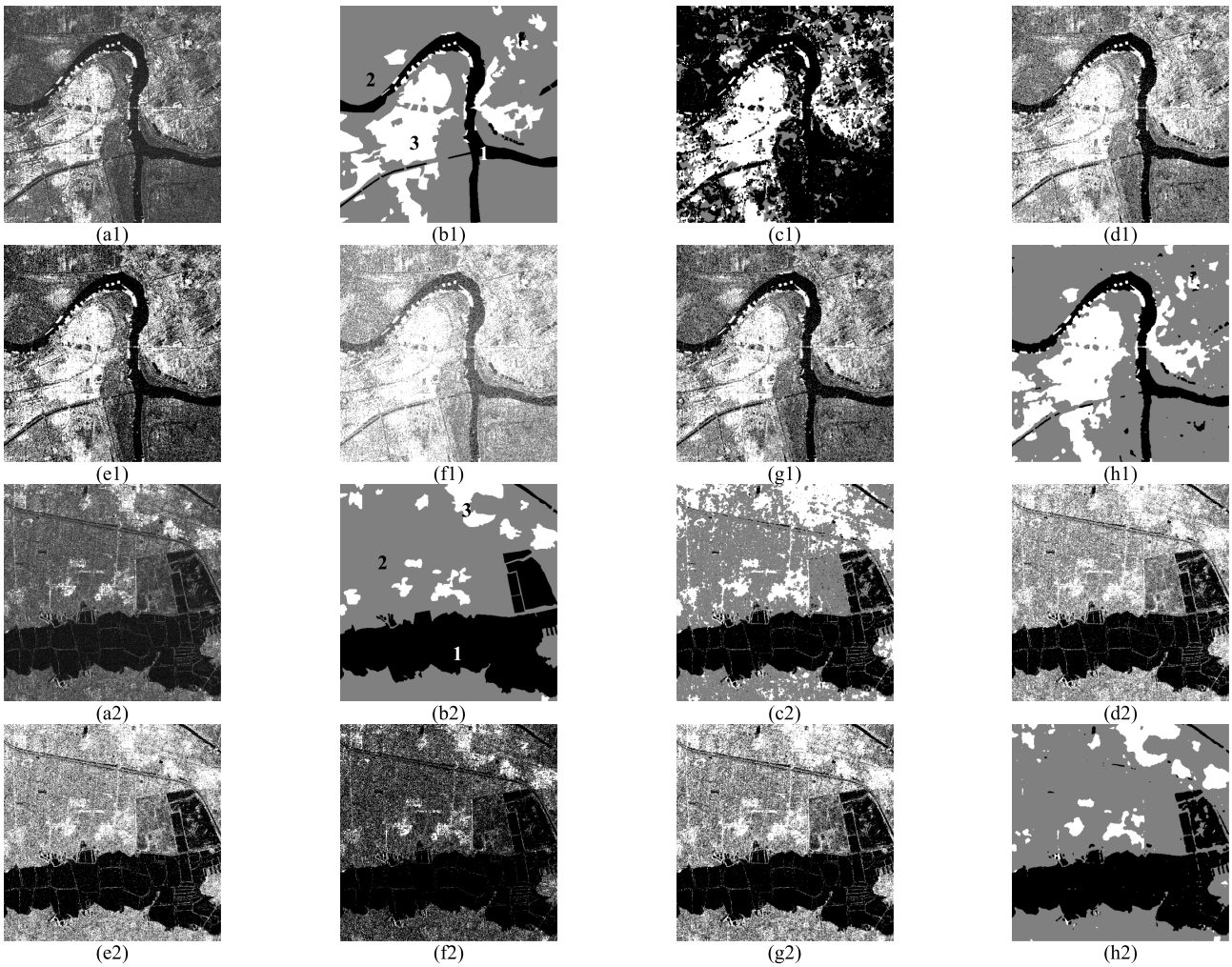
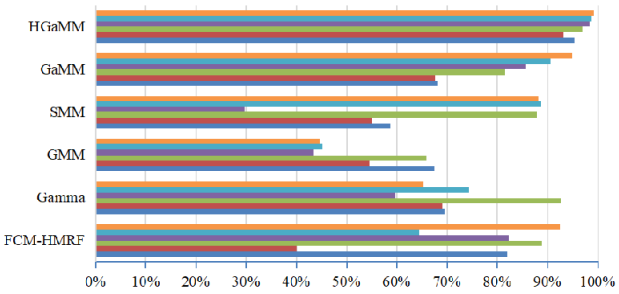


Fig. 11. Fourth experiment. (a1) and (a2) High-resolution SAR images. (b1) and (b2) Standard images. (c1) and (c2) FCM-HMRF algorithm. (d1) and (d2) Gamma algorithm. (e1) and (e2) GMM algorithm. (f1) and (f2) SMM algorithm. (g1) and (g2) GaMM algorithm. (h1) and (h2) Proposed algorithm.



	FCM-HMRF	Gamma	GMM	SMM	GaMM	HGaMM
Fig.10 (a1)	92.38%	65.12%	44.62%	88.15%	94.96%	99.19%
Fig.10 (a2)	64.45%	74.40%	45.13%	88.64%	90.58%	98.72%
Fig.10 (a3)	82.24%	59.73%	43.41%	29.68%	85.73%	98.39%
Fig.10 (a4)	88.81%	92.66%	65.91%	87.90%	81.52%	96.97%
Fig.11 (a1)	39.93%	69.02%	51.55%	51.97%	67.58%	93.19%
Fig.11 (a2)	82.00%	69.54%	67.39%	58.64%	67.95%	95.26%

Fig. 12. Accuracy evaluation of segmenting high-resolution SAR images.

The proposed algorithm yielded better segmentation results because it was able to model more accurately the complex distributions of spectral intensities using the weighted Gamma distributions as its components. Fig. 13 presents the curve fitting results of histograms for high-resolution SAR images in Figs. 10 and 11. The red and green curves present the fitting curves of the proposed HGaMM and its components, respectively. The histograms include heavy-tailed, asymmetric, peak, and flat features. The histograms of high-resolution SAR images are so flat, such as the right of histograms in Fig. 13(c), (e), and (f). Visually, the proposed algorithm was able to accurately fit the complex histograms, including the left small peak in Fig. 13(c) that corresponds to the black region in Fig. 10(h3). The right flat histogram was accurately fit using two components corresponding to regions 3 and 4 in Fig. 10(h3). For Fig. 11(a1) and (a2), spectral intensities are similar in regions 2 and 3, and boundaries are fuzzy between two regions. Hence, their histograms are heavy-tailed on the right. The proposed algorithm can accurately fit histograms and segment each region well. This shows that the proposed algorithm is able to fit complex distributions of pixels intensities for high-resolution SAR images and obtain accurate results.

Fig. 14 presents the point diagrams displaying the change in component number at each iteration. The initial value for Fig. 10(a1) and (a4) was set to be 8, while for

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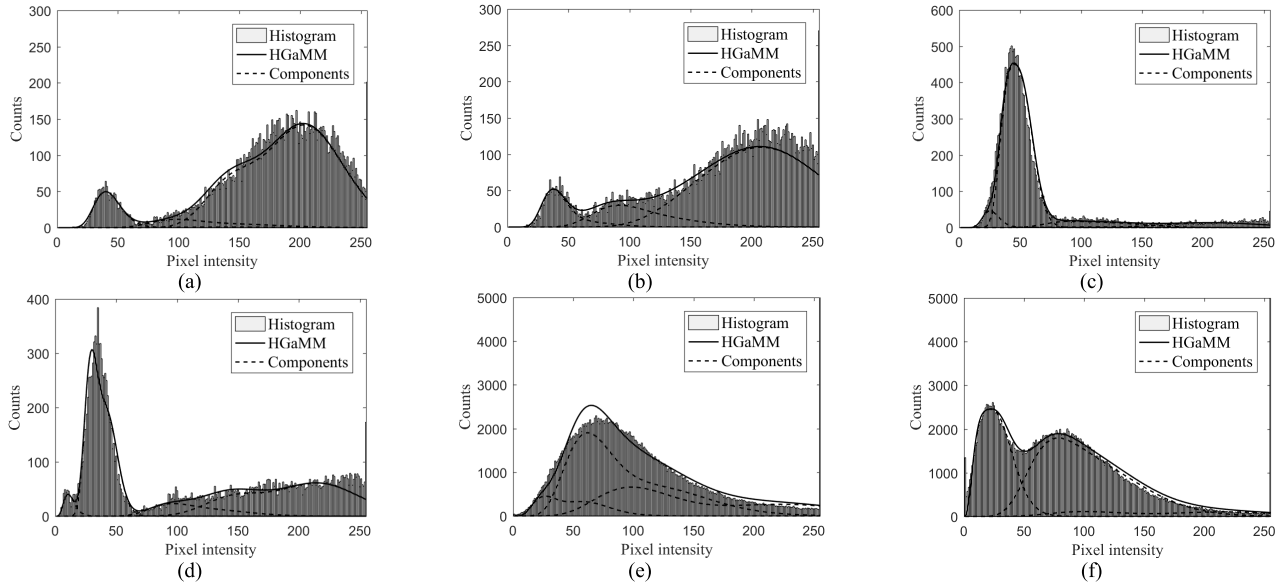


Fig. 13. Results of fitting histograms of SAR images with high resolution. (a)–(d) SAR images in Fig. 10(a1)–(a4). (e) and (f) SAR images in Fig. 11(a1) and (a2).

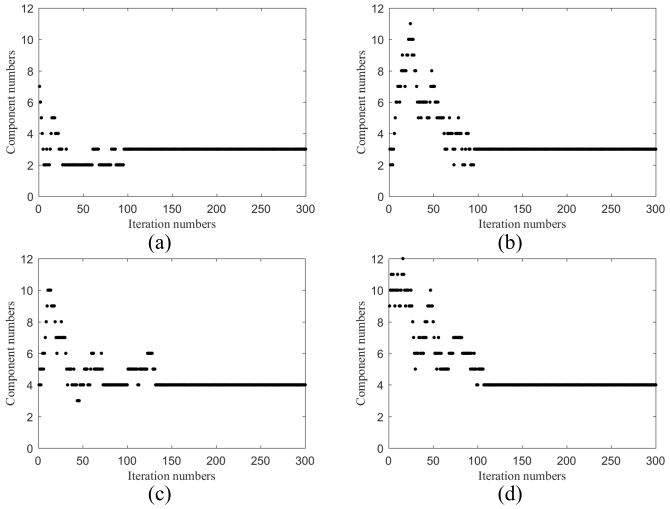


Fig. 14. Results of the component number changing with iteration. (a)–(d) SAR images in Fig. 10(a1)–(a4).

Fig. 10(a2) and (a3), the value was set to be 2. The component numbers for SAR images converged to 3, 3, 4, and 4 at about 100th, 100th, 140th, and 110th iterations in Fig. 14, respectively. The results matched the number of objective regions in SAR images, which suggests that the proposed algorithm can adaptively determine the optimum number of components. Moreover, the proposed algorithm is not affected by the initial value of the component number.

Table VIII lists the segmentation times of results for high-resolution SAR images. The average times of FCM-HMRF, Gamma, GMM, SMM, and GaMM and the proposed algorithms are 72.89, 1121.52, 164.28, 564.64, 1273.74, and 244.32 s, respectively. The time of the proposed algorithm was more than FCM-HMRF and GMM algorithms but less than Gamma, SMM, and GaMM algorithms. The efficiency of proposed algorithm is affected with the size of SAR images, and the times of Fig. 11(a1) and (a2) were more than other images, so are the other algorithms.

TABLE VIII
SEGMENTATION TIME OF RESULTS FOR SAR IMAGES

Algorithms	Time (s)					
	Fig. 10(a1)	Fig. 10(a2)	Fig. 10(a3)	Fig. 10(a4)	Fig. 11(a1)	Fig. 11(a2)
FCM-HMRF algorithm	32.94	34.30	32.49	36.90	142.64	158.05
Gamma algorithm	737.81	739.34	766.39	780.89	1827.48	1877.18
GMM algorithm	33.78	34.93	44.45	46.85	413.54	412.10
SMM algorithm	246.85	244.19	253.65	251.44	1174.42	1217.51
GaMM algorithm	762.79	752.17	832.76	829.03	2215.04	2250.61
The proposed algorithm	60.33	61.08	64.09	68.11	653.47	558.82

IV. CONCLUSION

To accurately segment high-resolution SAR images, we proposed a HGaMM-HMRF segmentation algorithm for high-resolution SAR images. Based on the results of numerous experiments, the conclusions can be summarized as follows.

The proposed algorithm is capable of building accurate statistical models and can accurately fit complex histograms, which yields better segmentation results for high-resolution SAR images. The algorithm incorporates the spatial relations of pixels to construct the weight of the component, aimed at improving the smoothness of noise and simplifying the solution for the component weight.

The results also show that the novel EM/ADMCMC was able to achieve accurate parameter estimation and adaptively

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obtain the optimum number of components. In the experiments, the number of elements for each component was set to be 2, which was sufficient to build the statistical model of spectral intensities. For more complex distributions of spectral intensities, the number can be set to 3. However, if the value is set to something extremely high, it would significantly affect efficiency. The main limitation of the proposed algorithm is that the coefficient η was set as constant in numerous experiments. However, by assessing the applicability of the proposed approach through experimentation, we found that the setting value is applicable in the segmentation of high-resolution SAR images.

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